Pairwise Guarantees for Directed Regularized Pearson-Type Classification

1 Setting and score

Let $\mathcal{H}_k$ be a real, separable RKHS with measurable feature map $\phi(z) = k(\cdot, z)$. We work with positive-semidefinite, pointwise nonnegative kernels satisfying

$$k(x, y) \geq 0, \qquad \sup_z k(z, z) \leq K < \infty, \qquad K > 0. \tag{1}$$

Nonnegativity implies

$$\|\phi(x) - \phi(y)\|^2_{\mathcal{H}_k} = k(x, x) + k(y, y) - 2k(x, y) \leq 2K, \tag{2}$$

which determines the concentration constants below.

Fix a reference distribution P_i and an incoming distribution Q. Let

$$Z_{i,1}, \ldots, Z_{i,n} \overset{\text{i.i.d.}}{\sim} P_i, \qquad Z'_1, \ldots, Z'_n \overset{\text{i.i.d.}}{\sim} Q,$$

with the two collections independent. Write

$$X_i = (Z_{i,1}, \ldots, Z_{i,n}), \qquad X' = (Z'_1, \ldots, Z'_n).$$

Define

$$\begin{aligned} \mu_P &= \mathbb{E}_{Z\sim P}\phi(Z), & T_P &= \mathbb{E}_{Z\sim P}[\phi(Z) \otimes \phi(Z)], \\ \widehat{\mu}_i &= \frac{1}{n}\sum_{r=1}^{n} \phi(Z_{i,r}), & \widehat{\mu}' &= \frac{1}{n}\sum_{r=1}^{n} \phi(Z'_r), \\ \widehat{T}_i &= \frac{1}{n}\sum_{r=1}^{n} \phi(Z_{i,r}) \otimes \phi(Z_{i,r}), & \widehat{\Delta}_i &= \widehat{\mu}' - \widehat{\mu}_i. \end{aligned}$$

Here $(u \otimes v)w = \langle v, w\rangle_{\mathcal{H}_k} u$, T_P is the uncentered second-moment operator, and $\Delta_i = \mu_Q - \mu_{P_i}$.

For fixed $\lambda > 0$, consider

$$\chi^2_\lambda(Q; P_i) := \frac{1}{2}\left\langle \Delta_i, (T_{P_i} + \lambda I)^{-1}\Delta_i \right\rangle_{\mathcal{H}_k}, \tag{3}$$

$$\widehat{\chi}^2_{\lambda,i} := \frac{1}{2}\left\langle \widehat{\Delta}_i, (\widehat{T}_i + \lambda I)^{-1}\widehat{\Delta}_i \right\rangle_{\mathcal{H}_k}. \tag{4}$$

This is a directed regularized RKHS Pearson-type discrepancy, with the reference distribution after the semicolon. It need not equal the classical Pearson divergence.

The kernel is fixed before sampling. A separate fixed kernel may be assigned to each reference, provided both windows in a comparison are evaluated using that reference's kernel. All guarantees below concern one fixed comparison and are over both sampled windows.

1.1 Concentration

For $\gamma \in (0,1)$, set

$$b_n(\gamma) := \sqrt{\frac{2K}{n}}\left(1+\sqrt{\log\frac{1}{\gamma}}\right), \qquad e_n(\gamma) := \frac{K}{\sqrt{n}}\left(1+\sqrt{\log\frac{1}{\gamma}}\right). \tag{5}$$

Lemma 1. *For every $\gamma \in (0,1)$,*

$$\mathbb{P}\left\{\left\|\widehat{\Delta}_i - \Delta_i\right\|_{\mathcal{H}_k} \le b_n(\gamma)\right\} \ge 1-\gamma, \tag{6}$$

$$\mathbb{P}\left\{\left\|\widehat{T}_i - T_{P_i}\right\|_{\mathrm{op}} \le e_n(\gamma)\right\} \ge 1-\gamma. \tag{7}$$

Proof. Independence and centering give

$$\begin{aligned}\mathbb{E}\left\|\widehat{\Delta}_i - \Delta_i\right\|_{\mathcal{H}_k}^2 &= \frac{1}{n}\left(\mathbb{E}_{Z\sim Q}k(Z,Z) - \|\mu_Q\|_{\mathcal{H}_k}^2\right)\\ &\quad + \frac{1}{n}\left(\mathbb{E}_{Z\sim P_i}k(Z,Z) - \|\mu_{P_i}\|_{\mathcal{H}_k}^2\right)\\ &\le \frac{2K}{n}.\end{aligned}$$

Thus the expected error norm is at most $\sqrt{2K/n}$. By (2), replacing one of the $2n$ observations changes that norm by at most $\sqrt{2K}/n$. McDiarmid's inequality therefore yields, for $t>0$,

$$\mathbb{P}\left\{\left\|\widehat{\Delta}_i - \Delta_i\right\|_{\mathcal{H}_k} - \mathbb{E}\left\|\widehat{\Delta}_i - \Delta_i\right\|_{\mathcal{H}_k} \ge t\right\} \le \exp\left(-\frac{nt^2}{2K}\right).$$

Taking $t = \sqrt{(2K/n)\log(1/\gamma)}$ proves (6).

For the operator, write $V(z) = \phi(z)\otimes\phi(z)$. Since $\|V(z)\|_{\mathrm{HS}} = k(z,z) \le K$,

$$\mathbb{E}\left\|\widehat{T}_i - T_{P_i}\right\|_{\mathrm{HS}}^2 = \frac{\mathbb{E}_{Z\sim P_i}\|V(Z)\|_{\mathrm{HS}}^2 - \|T_{P_i}\|_{\mathrm{HS}}^2}{n} \le \frac{K^2}{n}.$$

Moreover,

$$\|V(x)-V(y)\|_{\mathrm{HS}}^2 = k(x,x)^2 + k(y,y)^2 - 2|k(x,y)|^2 \le 2K^2.$$

Consequently, the norm $G = \left\|\widehat{T}_i - T_{P_i}\right\|_{\mathrm{HS}}$ has expectation at most $K/\sqrt{n}$ and replacement sensitivity at most $\sqrt{2}K/n$ for each of its n observations. McDiarmid's inequality gives

$$\mathbb{P}\{G - \mathbb{E}G \ge t\} \le \exp\left(-\frac{nt^2}{K^2}\right).$$

Taking $t = K\sqrt{\log(1/\gamma)/n}$ and using $\|\cdot\|_{\mathrm{op}} \le \|\cdot\|_{\mathrm{HS}}$ proves (7). □

2 Upper bound under the null hypothesis

Theorem 1 (Null upper bound)**.** *If $Q = P_i$, then for every $\rho_0 \in (0,1)$,*

$$\mathbb{P}\left\{0 \le \widehat{\chi}^2_{\lambda,i} \le U_n(\rho_0)\right\} \ge 1-\rho_0,$$

where

$$\boxed{U_n(\rho_0) := \frac{b_n(\rho_0)^2}{2\lambda} = \frac{K}{n\lambda}\left(1+\sqrt{\log\frac{1}{\rho_0}}\right)^2.} \tag{8}$$

Proof. Since $\widehat{T}_i \succeq 0$,

$$0 \le \widehat{\chi}^2_{\lambda,i} \le \frac{1}{2\lambda}\left\|\widehat{\Delta}_i\right\|_{\mathcal{H}_k}^2.$$

Under the null, $\Delta_i = 0$. Applying (6) at level ρ_0 proves the result. □

3 Lower bound under a separated alternative

3.1 Separation assumption

For a fixed pair, assume

$$\chi^2_\lambda(Q; P_i) \geq \delta > 0. \tag{9}$$

Alternatively, for a family $\mathcal{P}$ of admissible regimes, one may assume

$$\delta_{\min} := \inf_{\substack{P,Q\in\mathcal{P}\\ P\neq Q}} \chi^2_\lambda(Q; P) > 0$$

and take $\delta = \delta_{\min}$. Each score uses its reference kernel when kernels are reference-specific. Any novel distribution covered by this uniform guarantee must belong to $\mathcal{P}$.

Theorem 2 (Alternative lower bound)**.** *Under* (9)*, for every* $\rho_1 \in (0, 1)$*,*

$$\mathbb{P}\left\{\widehat{\chi}^2_{\lambda,i} \geq L_n(\delta, \rho_1)\right\} \geq 1 - \rho_1,$$

where

$$\boxed{L_n(\delta, \rho_1) := \frac{\left[\sqrt{2\delta} - b_n(\rho_1/2)/\sqrt{\lambda}\right]_+^2}{2\left(1 + e_n(\rho_1/2)/\lambda\right)}.} \tag{10}$$

Here $[x]_+ = \max\{x, 0\}$.

Proof. Set $b = b_n(\rho_1/2)$ and $e = e_n(\rho_1/2)$. By Lemma 1 and the union bound, with probability at least $1 - \rho_1$,

$$\left\|\widehat{\Delta}_i - \Delta_i\right\|_{\mathcal{H}_k} \leq b, \qquad \left\|\widehat{T}_i - T_{P_i}\right\|_{\mathrm{op}} \leq e.$$

Work on this event and write

$$A = T_{P_i} + \lambda I, \qquad \widehat{A} = \widehat{T}_i + \lambda I, \qquad c = 1 + \frac{e}{\lambda}.$$

Since $A \succeq \lambda I$,

$$\widehat{A} \preceq A + eI \preceq cA, \qquad \widehat{A}^{-1} \succeq c^{-1}A^{-1}.$$

Using the norm $\|v\|^2_{A^{-1}} = \langle v, A^{-1}v\rangle_{\mathcal{H}_k}$, the reverse triangle inequality gives

$$\begin{aligned}
\sqrt{2\widehat{\chi}^2_{\lambda,i}} &\geq \frac{1}{\sqrt{c}}\left\|\widehat{\Delta}_i\right\|_{A^{-1}} \\
&\geq \frac{1}{\sqrt{c}}\left[\|\Delta_i\|_{A^{-1}} - \left\|\widehat{\Delta}_i - \Delta_i\right\|_{A^{-1}}\right]_+ \\
&\geq \frac{1}{\sqrt{c}}\left[\sqrt{2\delta} - \frac{b}{\sqrt{\lambda}}\right]_+.
\end{aligned}$$

The last inequality uses (9) and $A^{-1} \preceq \lambda^{-1}I$. Squaring and dividing by 2 proves (10). □

3.2 Threshold interval

Accept the reference if $\widehat{\chi}^2_{\lambda,i} \leq \eta$ and reject it otherwise. Theorems 1–2 give the respective error bounds

$$\mathbb{P}_{H_0}\{\widehat{\chi}^2_{\lambda,i} > \eta\} \leq \rho_0, \qquad \mathbb{P}_{H_1}\{\widehat{\chi}^2_{\lambda,i} \leq \eta\} \leq \rho_1,$$

whenever

$$U_n(\rho_0) \leq \eta < L_n(\delta, \rho_1). \tag{11}$$

Here H_0 denotes $Q = P_i$, and H_1 denotes any fixed pair satisfying (9).

Writing

$$b_0 = b_n(\rho_0), \qquad b_1 = b_n(\rho_1/2), \qquad c = 1 + \frac{e_n(\rho_1/2)}{\lambda},$$

the interval is nonempty if and only if

$$\boxed{\delta > \frac{(b_1 + \sqrt{c}\, b_0)^2}{2\lambda}.} \tag{12}$$

Indeed, since $b_0 > 0$, the inequality $L_n(\delta, \rho_1) > U_n(\rho_0)$ is equivalent to

$$\sqrt{2\delta} > \frac{b_1 + \sqrt{c}\, b_0}{\sqrt{\lambda}}.$$

3.3 Threshold calibration and required separation

Choose a common error tolerance $\rho_0 = \rho_1 = \rho \in (0, 1)$ and set $\eta = U_n(\rho)$. In terms of the initial parameters,

$$\boxed{\eta = \frac{K}{n\lambda}\left(1 + \sqrt{\log\frac{1}{\rho}}\right)^2.} \tag{13}$$

This guarantees a null rejection probability at most ρ, independently of the separation assumption.

Substitution into (12) gives the separation required for a nonempty threshold interval:

$$\boxed{\begin{aligned}\delta > \frac{K}{n\lambda}\Bigg[1 + \sqrt{\log\frac{2}{\rho}} \\ + \left(1 + \sqrt{\log\frac{1}{\rho}}\right)\sqrt{1 + \frac{K}{\lambda\sqrt{n}}\left(1 + \sqrt{\log\frac{2}{\rho}}\right)}\Bigg]^2.\end{aligned}} \tag{14}$$

Under this condition, every fixed alternative pair with $\chi^2_\lambda(Q; P_i) \geq \delta$ satisfies

$$\mathbb{P}\left\{\widehat{\chi}^2_{\lambda,i} \leq \eta\right\} \leq \rho.$$

Condition (14) is equivalent to nonemptiness of the interval defined by our bounds. It is sufficient, but not necessary, for successful classification. These guarantees concern one fixed reference–incoming comparison; simultaneous control over the memory bank and repeated online decisions requires a separate analysis.